Holcombe Department of Electrical and Computer Engineering
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Lab Report
Experiment 7 Maximum Power Transfer Theorem

ECE 008

ECE-2080

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1) Abstract

In this seventh experiment, a circuit is provided. This report will cover the experimental process, as well as the theory and calculations required to prove, and apply the maximum power transfer theorem. These experimental results are significant because when applied to other circuits, they can find the conditions required to output the maximum power form a voltage source.

2) Introduction

The maximum power transfer theorem is foundational and is directly connected to real world applications. For this experiment, the circuit is constructed, and data is collected using a resistor box and digital multimeter. These are reinforced with hand calculations and compared. The aim of the lab is to prove the maximum power transfer theorem, and with these results, the voltage that transfers the maximum power can be found from the table created with the data. This is applicable in the real world for any application where power is desired, for example, speakers, and even in neuroscience for neural stimulators.

3) Theory and Procedure

Circuit Analysis requires understanding and application of the fundamental laws of circuits. The first being Ohm's Law (1). This law will be used to create variations of the power equation (2).

$$V = IR \tag{1}$$

Ohm's Law states that the voltage at a point in the circuit is equal to the current multiplied by the resistance. The next equation is the equation for power, which is defined in equation (2) as current times the voltage drop over the component power is being measured.

$$P = IV \tag{2}$$

Substituting Ohm's law into equation (2) allows for an equation written in terms of current and resistance. In terms of this lab, R is defined with a subscript L, to denote the resistance of the load.

$$P = I^2 R_L \tag{3}$$

The equation for current is defined as Thevenin voltage (R_{th}) divided by the Thevenin resistance (R_L) plus the resistance of the load (R_L). Thevenin resistance comes from removing the load desired to be measured. The Thevenin equivalent circuit can be seen in Figure (2). This equation will be plugged into the equation for power and evaluated to prove the equivalence of R_{th} and R_L which can be seen in Figure (1).

$$I = \frac{V_{th}}{R_{th}+R_L} \tag{4}$$

Proving that R_{th} and R_L are equivalent showcases that the maximum power transfer theorem. Finally, to compute the values for Power, a third and final iteration of the power equation is required. Also solved by plugging Ohm's law into the power equation, this equation relates voltage squared to resistance.

$$P = \frac{V^2}{R_L} \tag{5}$$

This completed the required equations for the completion of this lab.

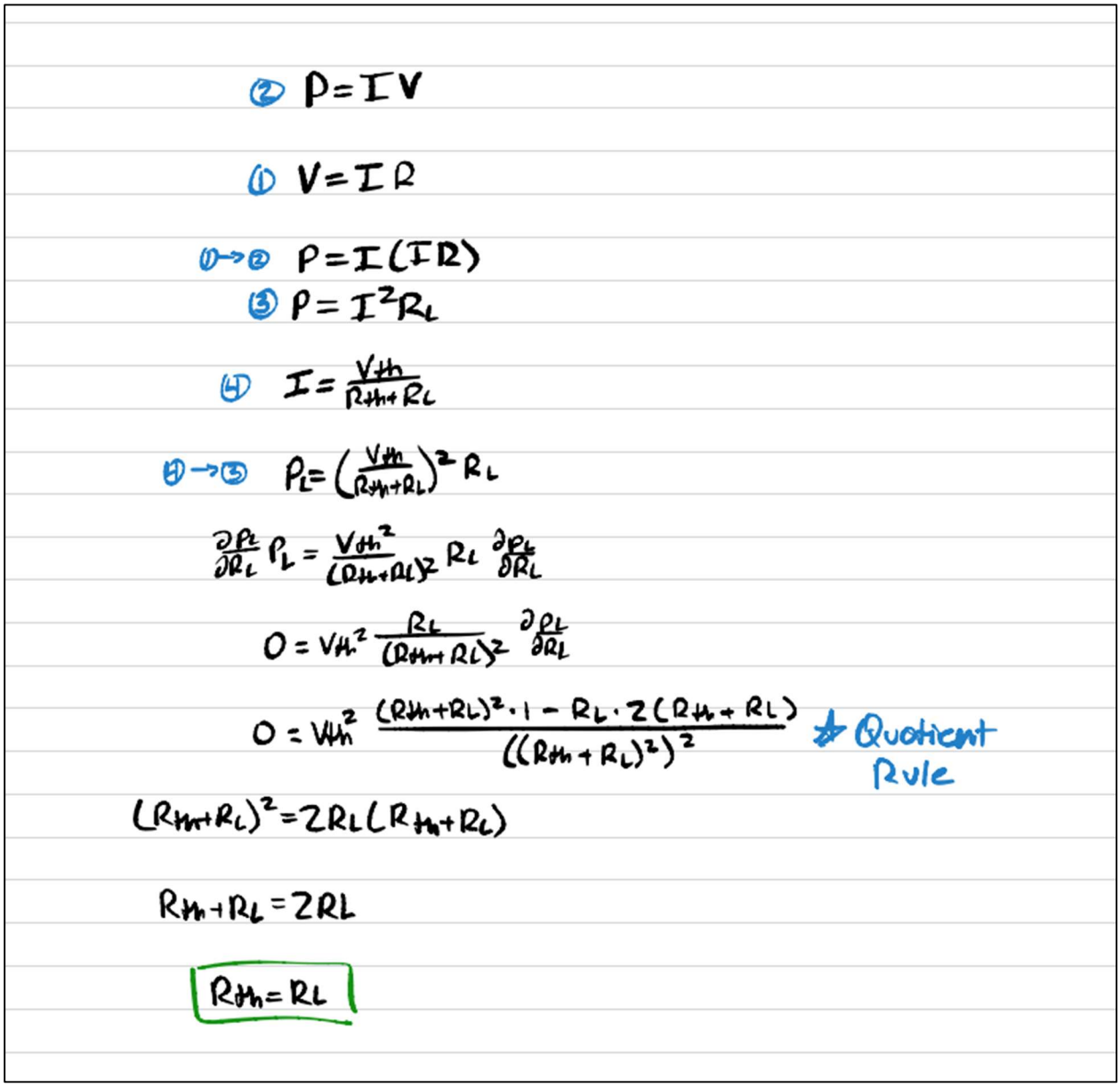


Figure 1: Algebraic Proof of Maximum Power Transport Theorem

I. Equations and handouts are received.

II. The equations are used to prove the Maximum Power Transfer Theorem shown above in Figure 1.

III. The Thevenin equivalent circuit is constructed and connected to the digital multimeter, voltage supply, and resistor box.

IV. Following the lab's safety procedures, once the circuit is verified to work, power is given to the bench, and the measurements can begin.

V. The resistor box is toggled to the correct resistance for the measurement, and the digital multimeter is used to record values of voltage drop over the loaded resistor.

VI. Hand calculations are used to solve the power using the resistance values and voltage from the experimental results using equation 3. These recorded values as well as calculated values can be found in Table 1.

4) Discussion of Results

The data was collected by keeping a constant 10V supply on the Thevenin equivalent of the circuit, which means that the load resistor was removed. The values for the rest of this experiment were recorded across the nodes that are now "open". This altered circuit is required to find the Thevenin resistance (R_{th}), which is utilized in equation 4. This altered circuit can be seen below in Figure 2.

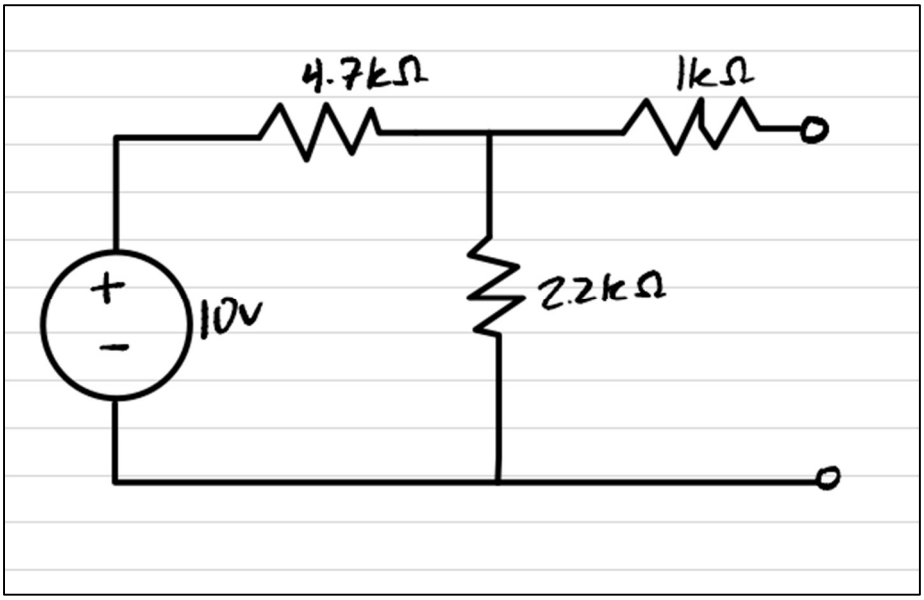


Figure 2: Thevenin Equivalent Circuit

With the Thevenin equivalent circuit made, the voltage was recorded as the resistance levels increased. This voltage was then calculated into power using equation 5.

The results from the experiment were exactly as the maximum power transfer theorem predicted. The theorem states that there is a given voltage where the power output will be the highest, and this was estimated experimentally by finding the maximum of the values. Looking at the data, there is such a point

where the power output is lower on both sides, and the value in the middle is the maximum power transfer. The voltage associated with this value of power is the desired result from this experiment. This value can be found from the experimental results in Table 1 below.

Table 1: Experimental Results

R_L $[K\Omega]$	V_L $[V]$	P_L $[mW]$
0.5	0.528	0.558
1.0	0.902	0.814
1.5	1.183	0.933
2.0	1.401	0.981
2.5	1.576	0.994
3.0	1.719	0.985
3.5	1.837	0.964
4.0	1.938	0.939
4.5	2.024	0.910
5.0	2.098	0.880

The values for power have a maximum at 0.944 mW, which means that this is the maximum. The values for resistance and voltage are noted because this combination produces the most power. This maximum value can also be seen visually when the values for power are plotted against the resistance. This can be seen below in Figure 3.



Figure 3: Experimental Results Plotted, Power v Resistance

The maximum of the graph occurs at 2.5 $K\Omega$, which lines up with table (1).

With all the data and results present, everything lines up. The theorem was mathematically proven, the constructed circuit returned proper values, and the data revealed a fixed voltage and resistance range where maximum power transfer occurs.

5) Conclusions

This experiment allows students to use a simple circuit, and tools found in the lab to prove and see a foundational theorem in action. Proving the equivalent resistances allows for an understanding at a theoretical level, but the construction of the circuit, data recording, and analyzation allows understanding at a physical level where the results can be visualized. Maximum power transfer theorem is a theorem with real world applications, and it is crucial to have an understanding. This experiment takes students through all the steps and allows them to practice, prove, and apply this important theorem.

References

[1] A. L. Duke and D. McAuliff, *Electrical Engineering Laboratory I*.

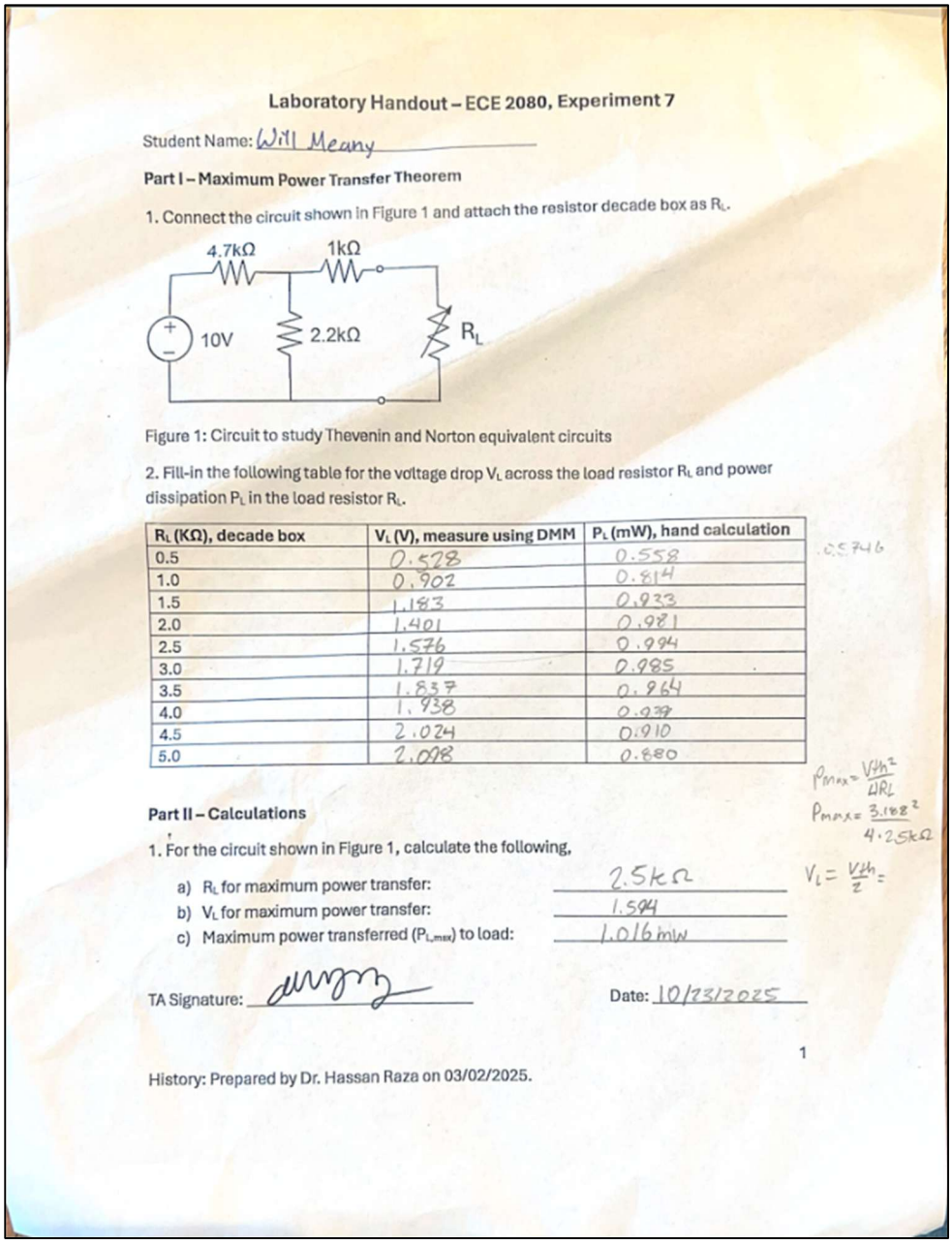
[2] D. K. Freeman *et al.*, "A Sub-millimeter, Inductively Powered Neural Stimulator," *Frontiers in Neuroscience*, vol. 11, Nov. 2017, doi: https://doi.org/10.3389/fnins.2017.00659.

[3] IEEE, "IEEE SA - The IEEE Standards Association - Home," *Ieee.org*, 2019. https://standards.ieee.org/

Appendix A: Technical Standards

[1] IEEE/ANSI 315A-1986, NFPA 70e

Appendix B: Documents



Laboratory Handout – ECE 2080, Experiment 7

Student Name: Will Meany

Part I – Maximum Power Transfer Theorem

1. Connect the circuit shown in Figure 1 and attach the resistor decade box as R_L.

Figure 1: Circuit to study Thevenin and Norton equivalent circuits

2. Fill-in the following table for the voltage drop V_L across the load resistor R_L and power dissipation P_L in the load resistor R_L.

R_L (KΩ), decade box	V_L (V), measure using DMM	P_L (mW), hand calculation
0.5	0.528	0.558
1.0	0.902	0.814
1.5	1.183	0.933
2.0	1.401	0.981
2.5	1.576	0.994
3.0	1.719	0.985
3.5	1.837	0.964
4.0	1.938	0.939
4.5	2.024	0.910
5.0	2.098	0.880

.5746

$P_{max} = \frac{V_{th}^2}{4R_L}$

$P_{max} = \frac{3.188^2}{4 \cdot 2.5k\Omega}$

$V_L = \frac{V_{th}}{2} =$

Part II – Calculations

1. For the circuit shown in Figure 1, calculate the following,

a) R_L for maximum power transfer: 2.5kΩ

b) V_L for maximum power transfer: 1.594

c) Maximum power transferred ($P_{L,max}$) to load: 1.016mW

TA Signature: [signature]

Date: 10/23/2025

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History: Prepared by Dr. Hassan Raza on 03/02/2025.

Figure 5: Lab Handout